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Ontology-based task planning for autonomous unmanned system: framework and principle

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Abstract. Operational knowledge is an important factor to improve the autonomy of unmanned systems. How to store and represent operational knowledge semantically and normatively, so that unmanned systems can autonomously accomplish task planning according to the high-level order of humans, carry out re-planning, and make decisions has been a new topic in the autonomous unmanned system field. This paper reviews several typical robot ontologies, as well as modeling and representation methods of uncertain information, proposes the concept of Ontology-based Task Planning(OTP) for the unmanned system, and introduces the principles of ontology construction. Finally, the application of ontology in unmanned combat systems are discussed.

Keywords. Ontology; Task planning; Artificial Intelligent planning; Unmanned system; Ontology-based task planning(OTP)

1. Introduction

At present, unmanned systems research is widely concerned in military and civilian fields, especially unmanned systems with autonomous capabilities[1–10].

The key to realizing autonomy is knowledge, such as knowledge of the battlefield environment, mission specifics, task specifics, and the knowledge of the unmanned system itself. How to store and represent the operation-related knowledge, enable unmanned systems to conduct task planning according to human's high-level task order, has become a new research field [11,12].

Based on the review of several ontology-based approaches for autonomy systems, uncertainty representation, and modeling methods, the concept of ontology-based task planning is proposed, and the principles of constructing task ontology are introduced. Finally, the challenges and research emphases of ontology-based task planning are discussed.

2. Concept of Ontology-based task planning for unmanned system

Ontology originates from the field of philosophy. The definition of Ontology by Studer in the field of knowledge engineering – “An ontology is a formal, explicit specification of a shared conceptualization.”[13,14] has been widely accepted. Ontology has the characteristics of clarity, formalization, and standardization. The knowledge representation using ontology technology has the advantages of being easy to share, clearly structured, accurate, and supporting reasoning.

Ontology-based task planning refers to unmanned systems conducting intelligently task planning based on ontology. It combines high-level symbolic planning and low-level motion planning such as trajectory



planning and path planning, indicating that unmanned systems may employ reasoning and inference based on the semantic represented task and scenario knowledge. Its framework is shown in Figure 1. Ontology-based task planning is a crossing field of knowledge engineering and task planning, and its connotation can be summarized as follows:

- Implement task planning according to knowledge. Ontology-based planning means that Unmanned systems can make use of experiential information when conducting task planning. These pieces of knowledge include human mission experience, experience learned by the unmanned system during the execution of previous missions, and experience learned in the simulated world. Ontology-based means that the task planning has the characteristics of growth ability and shareability, and can accomplish missions better according to experience just like human beings.
- It's an integrated planning framework of symbolic planning and motion planning. The three stages of mission planning(subtask sequence generation, task assignment, and trajectory/path planning) are highly coupled, which is more tightly coupled when we want autonomous unmanned systems to accomplish tasks cooperatively. OTP enables the combination of high-level task planning in the symbol domain and motion planning in the geometry domain.
- Promote information interaction from data to knowledge. Knowledge has two functions: Firstly, knowledge is more convenient in sharing among heterogeneous unmanned systems as well as human-machine interaction; Secondly, knowledge enables explanation and reasoning. All kinds of learning algorithms, which have received wide attention, are essentially learning some kind of knowledge. Such knowledge needs to be represented and interactive in a standardized way, so that it is convenient for all kinds of unmanned systems to query, share, and exploit. Only this way, the unmanned systems can adapt to the complex real-world and new environments never seen before.

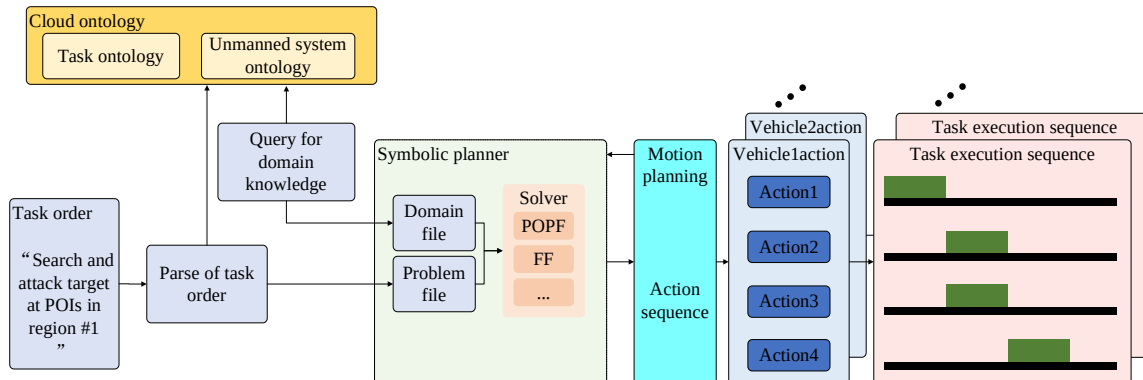


Figure 1. The framework of Ontology-based task planning

3. Related research

3.1. Ontology-based approaches for autonomous systems

Knowledge is the ladder to human wisdom and also an important basis for unmanned systems to improve their autonomous ability. The task planning of unmanned systems is not only a heavy workload but also requires the support of domain knowledge[15]. At present, the ontology approach has received extensive attention in the field of robotics, which has realized multiple robot ontologies and has been applied in the field of task planning.

RoboBrain is a super-large-scale robot knowledge graph[16]. This project uses an object-based database to store knowledge and is committed to realizing a knowledge engine for robot learning and knowledge sharing. Due to the large scale of the knowledge engine, its situation of practical application is not ideal. However, this project is one of the earliest research projects on knowledge ontology in the field of robotics.

KnowRob[17,18] is an action-centric ontology model. The action-centric organization of environmental knowledge enables unmanned systems easily access to the knowledge needed for their missions. The knowledge observed by the unmanned system is also added to the knowledge base, enabling the unmanned system to better adapt to a dynamically changing environment. KnowRob also uses Prolog to improve its knowledge reasoning ability[19].

RoboEarth is committed to establishing a worldwide Web for Robots to conduct knowledge interaction and learning for robots and to solve the problem of information reuse and interaction of heterogeneous robot systems[20]. A robot using RoboEarth queries the required resources according to the task algorithm at the beginning of the task and carries out task planning. If the current task has not been performed, the robot can record the learning process and upload the learned knowledge to the distributed knowledge base, which realizes knowledge sharing.

Reference [21] proposes a knowledge engine that provides spatial intelligence for robots. Spatial intelligence refers to spatial judgment ability. A robot with spatial judgment ability can visualize objects from different angles and Spaces, determine the relationship between objects, and deal with this spatial relationship. In task planning, the robot planner automatically processes spatial information and natural language instructions with the support of ontology and sensor data and no longer needs the help of task planning languages such as PDDL[22].

ORO (The Open Robots Ontology)[23,24] is object-oriented knowledge storage and reasoning platform. This system framework integrates data from different sources, such as sensors, domain knowledge, and human input information into a unified Ontology. Its research focuses on helping Robots to interact with humans and be able to use Pellet to carry out simple reasoning.

Literature [25] proposed an intelligent control architecture called Autonomous Marine Robots(AMR) for Marine Unmanned Underwater Vehicles (UUVs), which can dynamically plan and make a decision based on ontology so that it can adapt to complex environments.

Literature [26] applies ontology to the problem description, data interaction and sharing in the short-term mission planning of the space station. This research constructs the domain ontology of space station short-term mission planning, solves the challenge of problem description, data interaction and sharing, thereby effectively supporting the space station's short-term mission planning.

SWARMs[27] (Smart and Networking Underwater Robots in Cooperation Meshes) is dedicated to realizing the collaborative operation of Underwater robot systems produced by different manufacturers. It uses networked ontologies to solve the problem of exchanging heterogeneous information. As a kind of common information model, networked ontology represents knowledge of different domains and provides support for knowledge sharing within the system. Additionally, in SWARMs, the ontology supports different reasoning methods, such as ontology reasoning, rule-based reasoning, and multi-entity Bayesian Network (MEBN).

RTPO[11](Robot Task Planning Ontology) is a kind of indoor service-oriented robot task planning ontology. It uses OWL language to describe the domain knowledge of the robot and designs a task planning algorithm based on RTPO. RTPO is divided into three parts: task ontology, environment ontology, and robot ontology. Through the corresponding task planning algorithm, the robot can autonomously assign a high-level task.

Some features of these ontologies are shown in Table 1. In general, the application of knowledge ontology in unmanned systems mainly focuses on indoor service robots, and is still rare in military unmanned systems, especially in unmanned combat systems. Among all these ontologies, the performance of miniaturized and specialized ontologies is better than those of general large-scale ontologies, and OWL is the preferred representative language for ontology. Also, ontologies usually need an inference mechanism for knowledge reasoning.

3.2. Uncertainty knowledge modeling method

In the real world, especially on the battlefield, an unmanned system must cope with a lot of uncertain information. So, Uncertain knowledge modeling is very important. At present, the mathematical theories dealing with uncertain knowledge mainly include fuzzy logic, Bayesian network, Markov logic network, etc.

Literature [28] introduced a method of realizing fuzzy ontology through OWL2. Probability annotation was used to represent the uncertainty of information and fuzzy ontology elements such as fuzzy data types, fuzzy concepts, and fuzzy modifiers were represented.

Table 1. Comparison of typical robotic ontologies

Name	Scale	Application field	Represent method	Knowledge inference
RoboBrain	very large scale	Service robot	Graph structure	Robot Query Language (RQL)
KnowRob	small	Service robot	Web OWL	Rapyuta
RoboEarth	large scale	Service robot	OWL	Rapyuta
Knowledge Engine in reference[20]	small	Service robot	OWL	HermiT
ORO	small	Robot community	OWL	Pellet
RTPO	small	Service robot	OWL	SWI-prolog
AMR	small	Unmanned underwater vehicle	OWL	--
Task ontology for space station	small	space station	OWL	--
SWARMS	small	Marine unmanned equipment	PR-OWL	Pellet

Literature[29] proposes a Fuzzy Ontology Construction Method – FODM and constructs a fuzzy ontology for the submarine environment.

Multi-Entities Bayesian Network (MEBN) [30] is an extension of the Bayesian Network, which uses entity sets and various relationships among entities to represent the uncertainties in the environment. PR-OWL is realized based on MEBN theory. It uses a probabilistic annotation method to express uncertain information, which can effectively solve reasoning problems. It has been applied in smart manufacturing systems[31], maritime early warning[32], and other fields.

3.3. Role of ontology in task planning of unmanned systems

The main functions of ontology in the task planning of unmanned systems are as follows:

- First, it supports intelligent task planning. The knowledge ontology can store planning domain knowledge or combat rules that can be applied in missions. Besides, with the support of ontology, the unmanned system can automatically generate subtask sequences according to the high-level order from humans and check whether the task can be achieved or not. Large-scaled heterogeneous unmanned systems on the battlefield in the future will inevitably require an efficient task planning and management mechanism to realize efficient and intelligent task planning.
- Second, Supporting knowledge sharing. Ontology provides a basis for knowledge sharing among heterogeneous unmanned systems, standardizes information exchange, and enables knowledge shared through the network. Through the accumulation and sharing of operational knowledge, the intelligence of unmanned systems will be enhanced.

- Thirdly, supporting knowledge reasoning. Ontology is a structured description of concepts and their relations. It enables unmanned systems to reason, deal with uncertainty and improve their ability in complex environments.

4. Principles of task ontology construction

The construction of task ontology needs to consider many factors. This section explains three main aspects: application scope and goals, non-functional requirements, and functional requirements.

4.1. Clarify the application scope of the ontology

The task ontology should base on their application scenarios, and avoids the problem that the ontology is too generalized and too large to be applied. The task ontology should accomplish two main tasks. One is to clearly define the concepts related to domain knowledge. Secondly, defines the relationship between concepts. The ontology should be able to describe heterogeneous unmanned systems (such as sensors, actuators, and other components), their working environment (such as terrain, threats, obstacles, and other battlefield elements), and the information that must be exchanged between unmanned systems.

4.2. Non-functional requirements of the task ontology

Interoperability, modularity, reusability, and extensibility are the non-functional characteristics of an ontology. Interoperability is the basis for information exchange. Modularity requires the decoupling of ontology elements, which is conducive to a flexible combination of ontology elements. Reusability enables each element of ontology to be easily applied to different fields or to adapt to new scenarios through simple modification. Extensibility is a fundamental requirement for ontology capabilities to grow and adapt to more complex scenarios.

4.3. Functional requirements

- The ontology should contain the task planning model. Supporting task planning is the primary functional requirement of an ontology. Firstly, the ontology should be able to describe the exact actions, preconditions, and effects of actions. Secondly, it should describe the scenarios of unmanned systems operated in, which is convenient for knowledge sharing, reasoning and task collaboration. Figure 2 shows the relationships between the major domains of a task ontology.

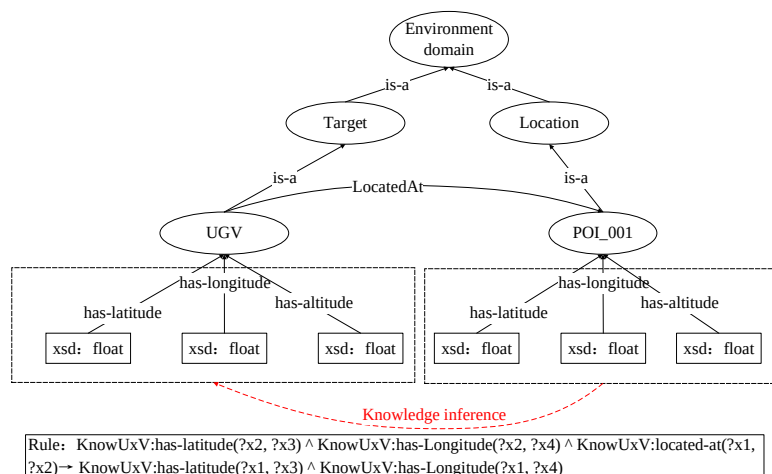


Figure 2. Diagram of knowledge inference

- The ontology should clearly define the category and attribute capability of unmanned systems. Attributes and capabilities of unmanned systems are important constraints of task planning. When parsing high-level task orders, unmanned systems need to query their attributes and capabilities in ontology to see if they meet the conditions of task execution.

- Ontologies should be able to support battlefield modeling. Battlefield information is an important guarantee for the correct task execution of an unmanned combat system. The ontology should define and represent altitude, electromagnetic radiation, wind speed, weather, threat, target, etc. In the marine environment, it should also include salinity, temperature, ocean currents, etc. Besides, the ontology should contain the sensor model and use the sensor information to reconstruct the physical environment.
- The ontology should be able to support the representation and inference of uncertain information. A battlefield environment is a dynamic environment, which contains a lot of uncertainty. The task ontology should be able to label this uncertain information and support knowledge reasoning based on context data.

5. Conclusion

This paper reviews the application of ontology in task planning, proposed the concept of Ontology-based Task Planning(OTP) for unmanned systems, explained the principles of ontology construction. Knowledge can support high-level task planning and decision-making, improve adaption to complex environments, improve autonomy and reduce human intervention. At present, ontology technology has been applied in simple task planning, knowledge reasoning, and sharing, but more research results are expected for 1)more complex task planning ability and 2) knowledge representation and learning in unstructured and open environments. Also, 3)the uncertain information representation caused by the uncertainty of the battlefield environment, and 4)the problem of information fusion and reasoning are areas that need to be paid more attention to. To sum up, the ontology-based approach is a potential research direction for achieving autonomy.

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